

RSA cryptosystem

The RSA cryptosystem involves using a public encryption key of a modulus $n = pq$, where p and q are large primes and an public exponent e that is relatively prime to $(p-1)(q-1)$:

$$\text{GCD}(e, \phi(n)) = 1$$

$$\text{GCD}(e, \phi(pq)) = 1$$

$$\text{GCD}(e, (p-1)(q-1)) = 1$$

To encrypt a message $C = M^e \bmod n$ is used where C is the ciphertext and M is the plaintext.

The exponent e and modulus n are the public keys.

To decrypt a message $M = C^d \bmod n$ is used where d is the private decryption exponent.

To obtain d , knowledge of the factorization of n must be known.

Example:

An RSA cryptosystem uses a modulus of $n=(43)(59)=2537$ and an encryption exponent $e=11$.

To encrypt the message "math" we translate into a numerical equivalent using $\mathbb{Z}_{26}$: 12 00 19 07

The modulus can accommodate a max block size of 4-digits. Encrypting by 4-digit blocks:

$$C = 1200^{11} \bmod(2537) = 0647$$

$$1200^2 \bmod 2537 = 1521$$

$$1200^4 \bmod 2537 = 2234$$

$$1200^8 \bmod 2537 = 477$$

$$1200^{10} \bmod 2537 = 2472$$

$$1200^{11} \bmod 2537 = 647$$

$$C = 1907^{11} \bmod(2537) = 2267$$

$$1907^2 \bmod 2537 = 1128$$

$$1907^4 \bmod 2537 = 1347$$

$$1907^8 \bmod 2537 = 454$$

$$1907^{10} \bmod 2537 = 2175$$

$$1907^{11} \bmod 2537 = 2267$$

The encrypted message is: 06 47 22 67

To decrypt we compute the inverse of e modulo $((p-1)(q-1))$.

In this case we need the inverse of 11 modulo $(43-1)(59-1) = 2436$.

The Extended Euclidean Algorithm gives:

	Quotient	Remainder	s	t
$2436 \bmod 11 = 5$	221	5	1	-221
$11 \bmod 5 = 1$	2	1	$-2(1) = -2$	$1 - 2(-221) = 443$
$5 \bmod 1 = 0$				

We have $\text{GCD}(2436, 11) = 1$ and $-2(2436) + 443(11) = 1$. So 443 is an inverse of 11 modulo 2436.

Using $d=443$ we compute:

$$M = 0647^{443} \bmod 2537 = 1200$$

$$M = 2267^{443} \bmod 2537 = 1907$$

The decrypted message is 12 00 19 07 which corresponds to the letters "math".

Note: If you are using a calculator for this, use fast modular exponentiation and successive squaring:

$0647^2 \bmod 2537 = 4$
 $0647^4 \bmod 2537 = 16$
 $0647^8 \bmod 2537 = 256$
 $0647^{16} \bmod 2537 = 2111$
 $0647^{32} \bmod 2537 = 1349$
 $0647^{64} \bmod 2537 = 772$
 $0647^{128} \bmod 2537 = 2326$
 $0647^{256} \bmod 2537 = 1392$
 $0647^{256+128=384} \bmod 2537 = 580$
 $0647^{384+32=416} \bmod 2537 = 1024$
 $0647^{416+16=432} \bmod 2537 = 140$
 $0647^{432+8+2=442} \bmod 2537 = 1288$
 $0647^{442+1=443} \bmod 2537 = 1200$

$2267^2 \bmod 2537 = 1864$
 $2267^4 \bmod 2537 = 1343$
 $2267^8 \bmod 2537 = 2379$
 $2267^{16} \bmod 2537 = 2131$
 $2267^{32} \bmod 2537 = 2468$
 $2267^{64} \bmod 2537 = 2224$
 $2267^{128} \bmod 2537 = 1563$
 $2267^{256} \bmod 2537 = 2375$
 $2267^{256+128=384} \bmod 2537 = 494$
 $2267^{384+32=416} \bmod 2537 = 1432$
 $2267^{416+16=432} \bmod 2537 = 2118$
 $2267^{432+8+2=442} \bmod 2537 = 848$
 $2267^{442+1=443} \bmod 2537 = 1907$

Exercises:

1. An RSA cryptosystem has public key $\{n=319, e=17\}$ and a block size of 3-digits.
 - a) Encrypt the number 123.
 - b) Determine the decryption exponent d .
 - c) Decrypt the number 227.

2. An RSA cryptosystem has public key $\{n=2537, e=13\}$ and a block size of 4-digits.
 - a) Encrypt the plaintext "stop".
 - b) Determine the decryption exponent d .
 - c) Decrypt the ciphertext "09 81 04 61".
 - d) Decrypt the ciphertext "06 67 19 47 06 71".

Answers:

1.

- a) $123^2 \bmod 319 = 136$
 $123^4 \bmod 319 = 313$
 $123^8 \bmod 319 = 036$
 $123^{16} \bmod 319 = 020$
 $123^{17} \bmod 319 = 227$
The encrypted number is 227

- b) $n = (11)(29)$
 $\phi(n) = 280$

	Q	R	s	t
$280 \bmod 17 = 8$	16	$280 - 16(17) = 8$	1	-16
$17 \bmod 8 = 1$	2	$17 - 2(8) = 1$	$-2(1) = -2$	$1 - 2(-16) = 33$
$8 \bmod 1 = 0$				

We have $\text{GCD}(280, 17) = 1$ and $-2(280) + 33(17) = 1$.
The decryption exponent is $d = 33$.

- c)
- $227^2 \bmod 319 = 170$
 $227^4 \bmod 319 = 190$
 $227^8 \bmod 319 = 053$
 $227^{16} \bmod 319 = 257$
 $227^{32} \bmod 319 = 016$
 $227^{33} \bmod 319 = 123$
The decrypted number is 123.

2.

- a) 20 81 21 82
b) $d = 937$
c) "help"
d) "silver"